

SONiX 32-Bit Cortex-M0 Micro-Controller

CMP Example Code

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AMENDENT HISTORY

Version	Date	Description
1.0	2024/11/26	1. First version.

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1 OVERVIEW

This document provides our customers the C program examples of RAIL TO RAIL COMPARATOR.

2 FEATURE

- Rail-to-Rail structure comparators.
- Use ARM-compiler V6.12

3 HW

- The starter kit board of vendor desired MCU.
- Connect CMnP0 / CMnN0 / OP00 with desire voltage source.

4 EXAMPLE CODE

4.1 Initialization

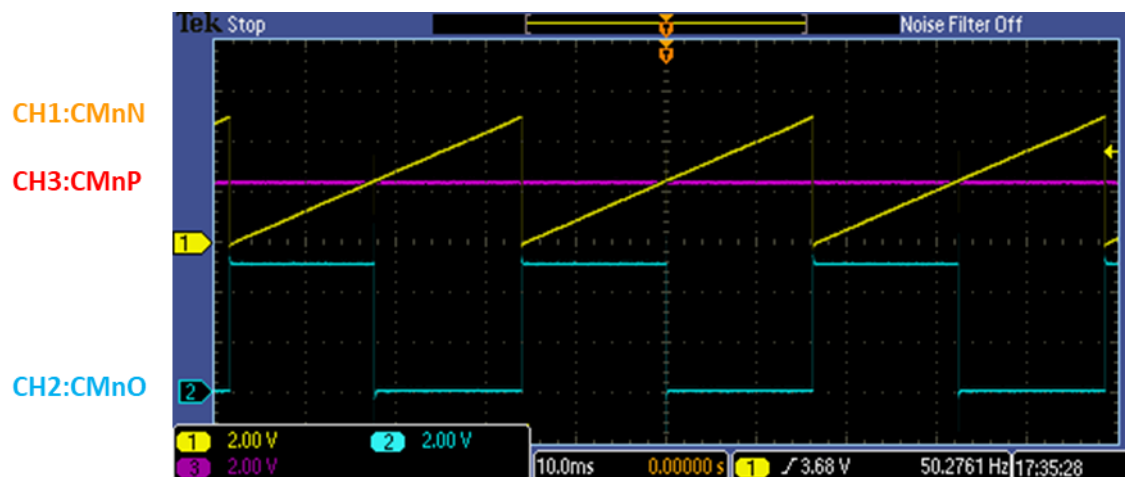
Initial CMP function. User can modify the settings of CMP in this function. Depending on the chip series, the source selection for the positive and negative input functions may be different.

```
65 //-----
66 //User Code starts HERE!!!
67 CMP_init();
```

```
88 /*****
89 * Function      : CMP_init
90 * Description    : Initialization of CMP0/CMP1/CMP2/CMP3
91 * Input         : None
92 * Output        : None
93 * Return        : None
94 * Note          : None
95 *****/
96 void CMP_init(void)
97 {
98     __CMP_ENABLE_CMPHCLK;
99
100 #if CMP_VIREF_ENABLE
101     __ADC_ENABLE_HCLK
102     CMP_VIREF_Enable(0x7F);
103 #endif
104
105 #if CMP0_ENABLE
106     SN_CMP->CTRL |= (mskCMP_CM0PS_VIREF0 | mskCMP_CM0O_ENABLE | mskCMP_CM0G_FALLING);
107     SN_CMP->IE |= mskCMP_CM0IE_ENABLE;
108     NVIC_EnableIRQ(CMP0_IRQn);
109     __CMP0_ENABLE;
110 #endif
111
112 #if CMP1_ENABLE
113     SN_CMP->CTRL1 |= (mskCMP_CM1PS_VIREF0 | mskCMP_CM1O_ENABLE | mskCMP_CM1G_FALLING);
114     SN_CMP->IE |= mskCMP_CM1IE_ENABLE;
115     NVIC_EnableIRQ(CMP1_IRQn);
116     __CMP1_ENABLE;
117 #endif
118
119 #if CMP2_ENABLE
120     SN_CMP->CTRL1 |= (mskCMP_CM2PS_VIREF0 | mskCMP_CM2O_ENABLE | mskCMP_CM2G_FALLING);
121     SN_CMP->IE |= mskCMP_CM2IE_ENABLE;
122     NVIC_EnableIRQ(CMP2_IRQn);
123     __CMP2_ENABLE;
124 #endif
125
126 #if CMP3_ENABLE
127     SN_CMP->CTRL1 |= (mskCMP_CM3PS_VIREF0 | mskCMP_CM3O_ENABLE | mskCMP_CM3G_FALLING);
128     SN_CMP->IE |= mskCMP_CM3IE_ENABLE;
129     NVIC_EnableIRQ(CMP3_IRQn);
130     __CMP3_ENABLE;
131 #endif
132 }
```

5 RESULT

The results of CMP can be observed through CMnP0 / CMnN0 / OP0O pins. Comparator compares positive terminal's voltage and negative terminal's voltage, and then output result to output pin. When $V+ > V-$, comparator outputs high status. When $V+ < V-$, comparator outputs low status.



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